

Upper & Lower Respiratory Disorders

Conducting Passages

Upper respiratory tract

Nasal cavity

Pharynx

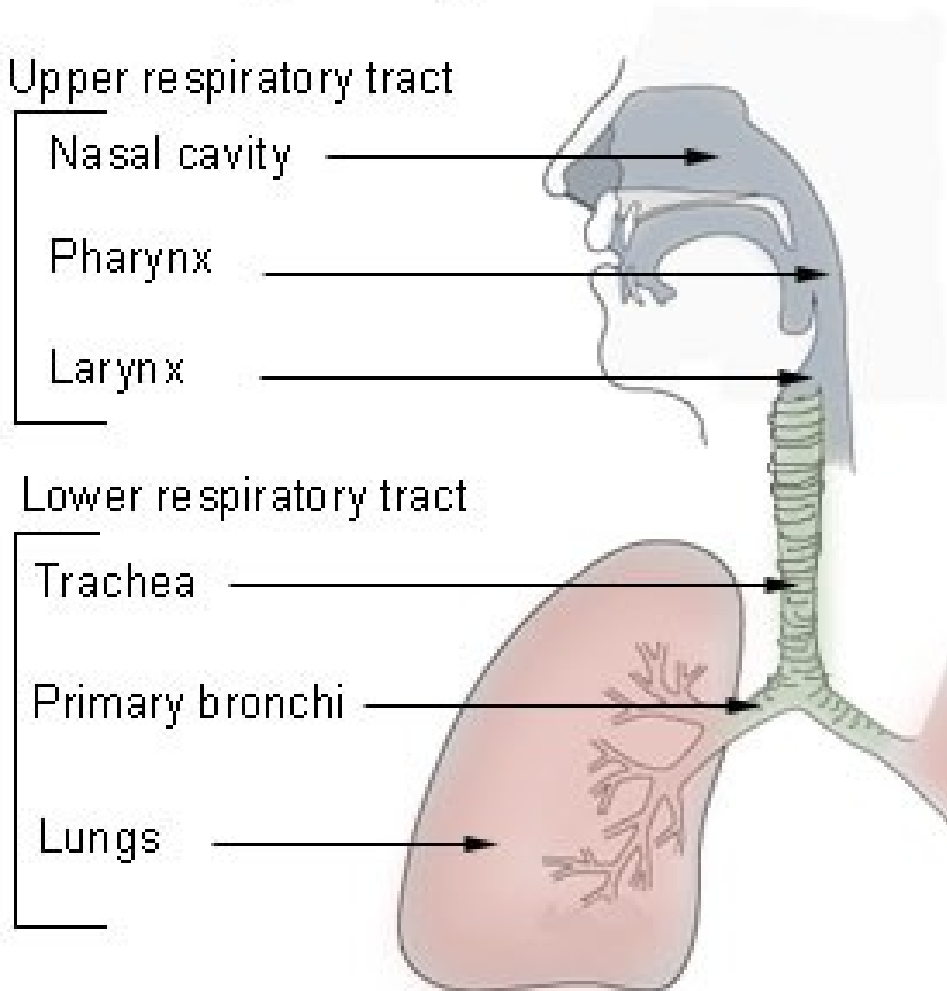
Larynx

Lower respiratory tract

Trachea

Primary bronchi

Lungs

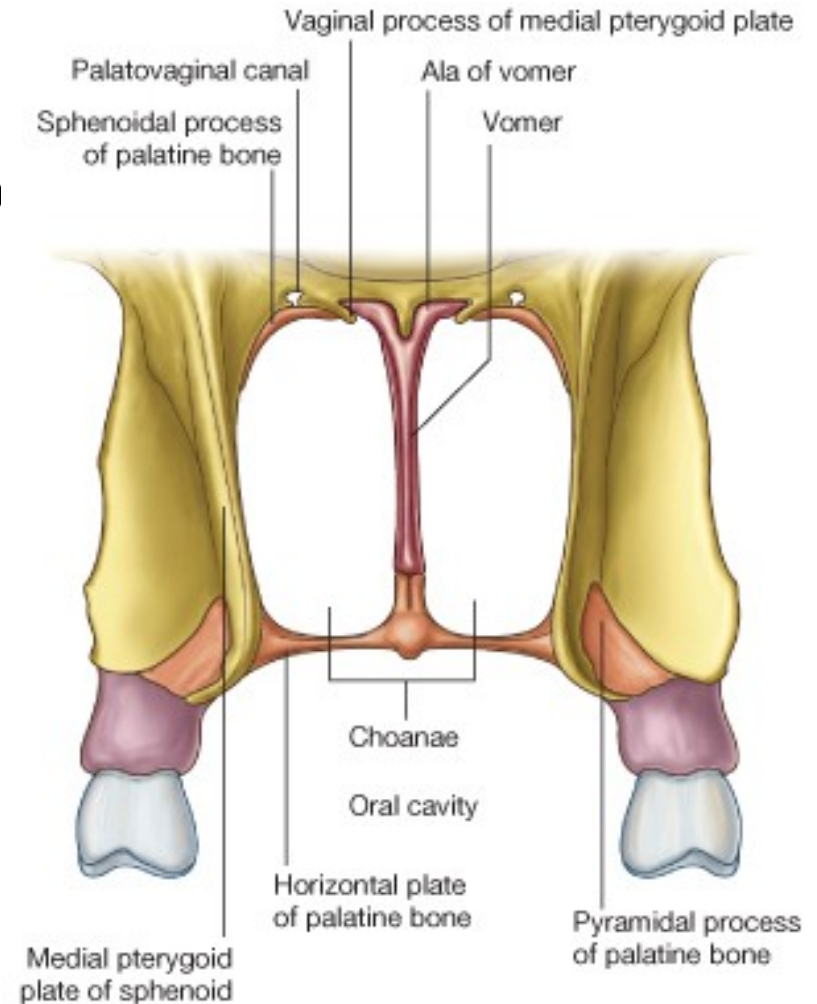


DISEASES OF THE UPPER RESPIRATORY TRACT (nose, nasopharynx, & larynx)

Congenital Diseases

Choanal atresia

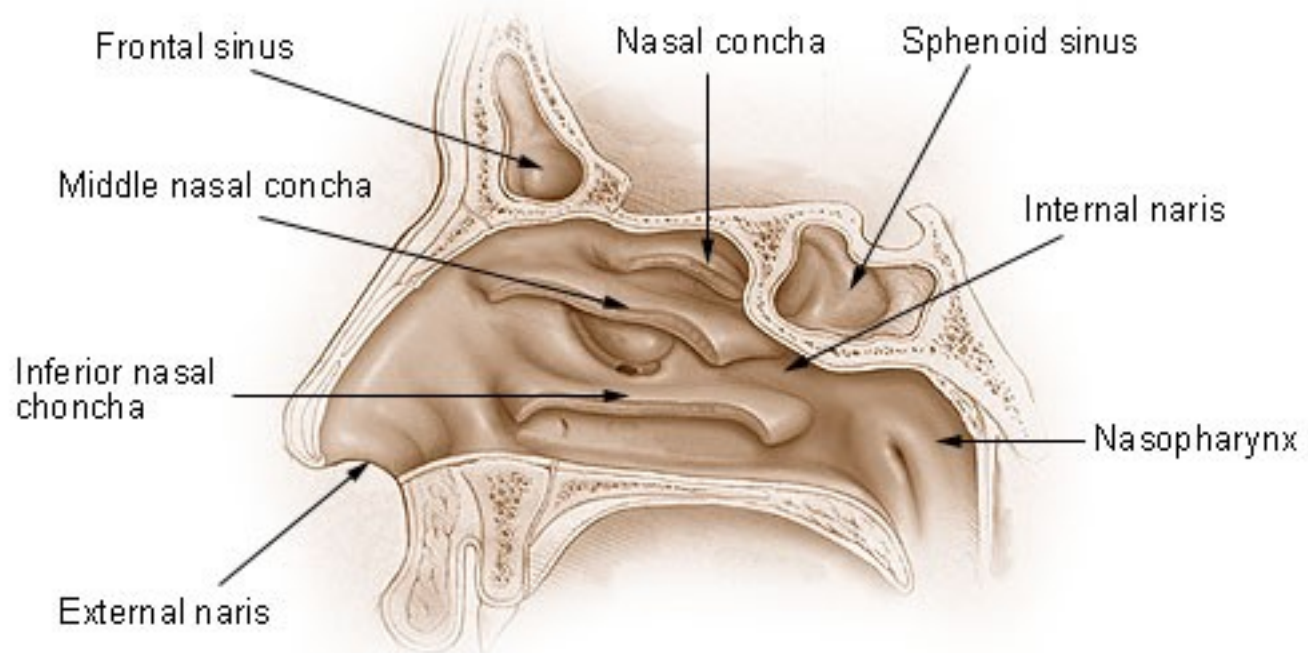
- Unilateral or bilateral bony septum between the nose and the pharynx
- Newborn turns cyanotic when breast-feeding.
 - Crying causes the child to "pink up" again.



Normal Anatomy

Nose

Nose and Nasal Cavities



Inflammation

Infectious rhinitis

- “common cold”
- Most cases are caused by viruses

- The most common pathogens are rhinoviruses, others include coronaviruses, respiratory syncytial viruses, parainfluenza virus , adenoviruses & enteroviruses.

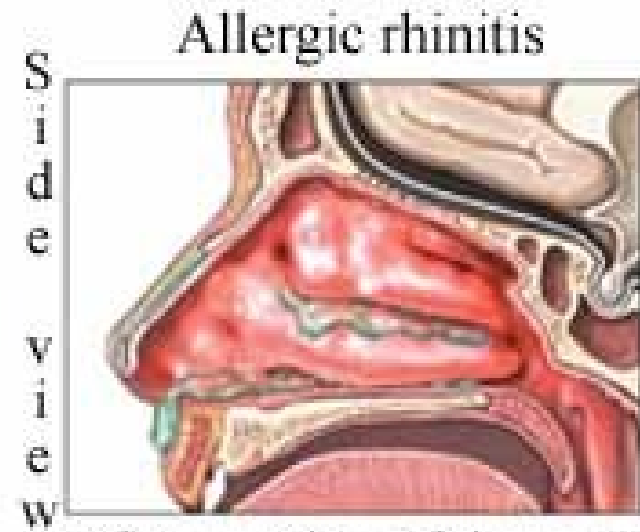
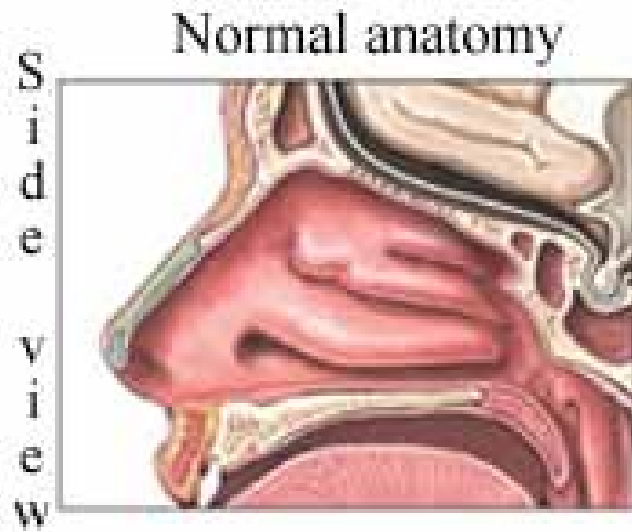
- The clinical features are nasal congestion accompanied by watery discharge; sneezing; scratchy, dry sore throat; and a slight increase in temperature that is more pronounced in young children

Morphology

- The nasal mucosa is thickened, edematous & red , the nasal cavities are narrowed & the turbinates are enlarged.
- Bacterial superinfection results in mucopurulent or suppurative exudate.

Allergic rhinitis (Hay fever)

- An allergic reaction with immunoglobulin E – mediated immune reaction triggered by allergens such as pollen.
- The allergic reaction is characterized by marked mucosal edema, redness & mucus secretion accompanied by leukocytic infiltration (mostly eosinophils)



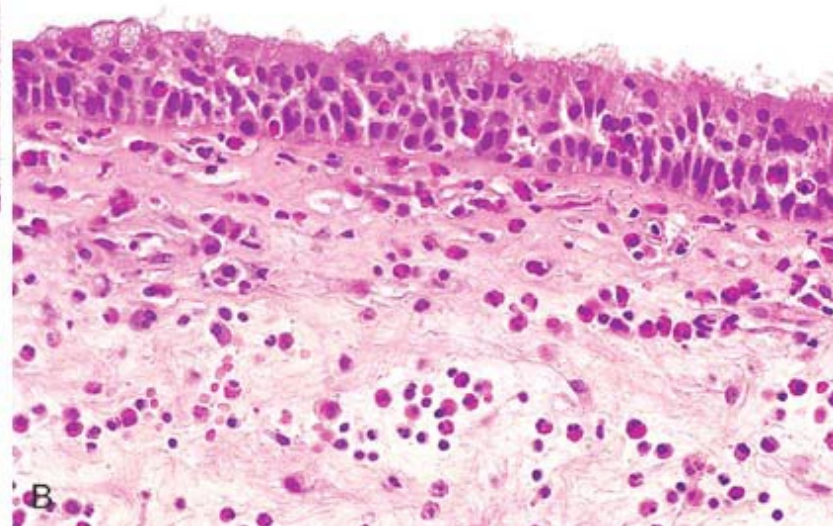
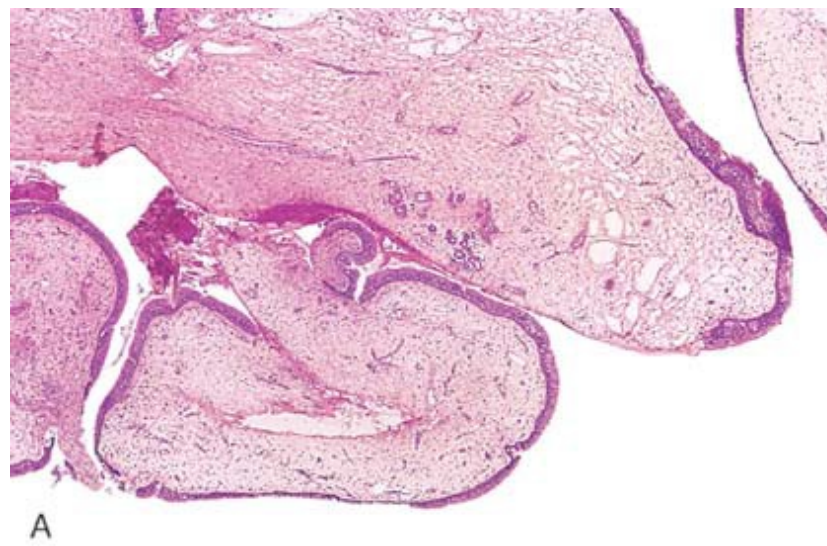
Inflammation of the nasal mucus layer with exudate in the airway.

Nasal polyp

- Focal protrusions of the nasal mucosa due to recurrent attacks of rhinitis
- May reach up to 3 to 4 cm in length

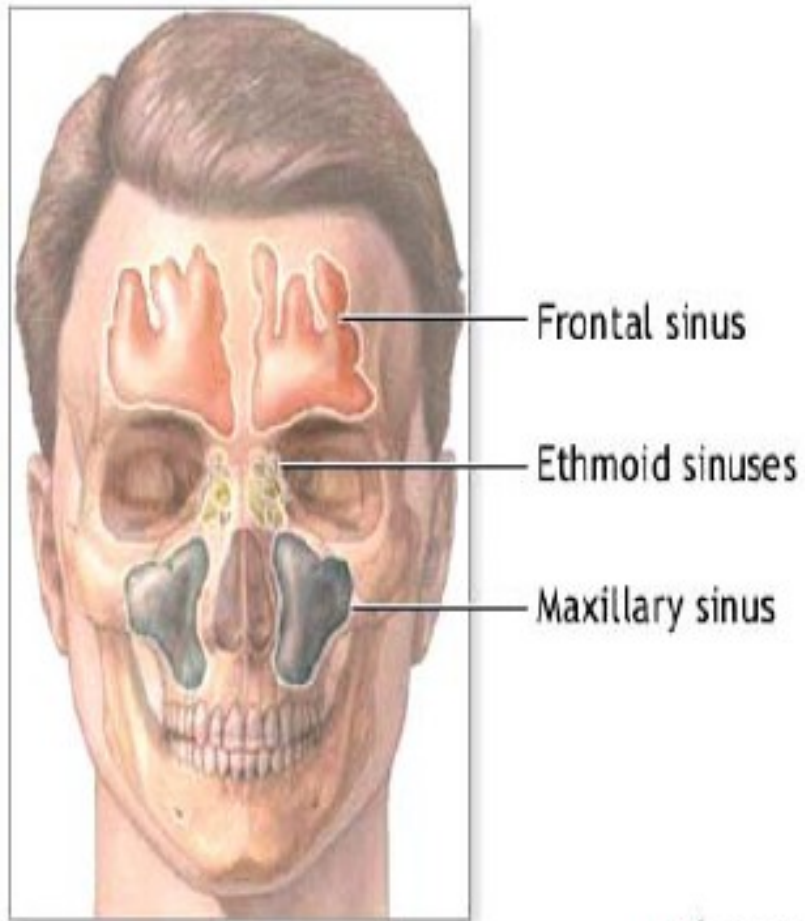


- On histologic examination , polyps consist of edematous mucosa having loose stroma with mucous glands & infiltrated by inflammatory cells



Chronic rhinitis

- It is sequel to repeated attacks of acute rhinitis whether allergic or microbial in origin with superimposed bacterial infection.



Sinusitis

- Most often preceded by acute or chronic rhinitis, but it can arise by extension of a periapical infection through the bony floor of maxillary sinus.

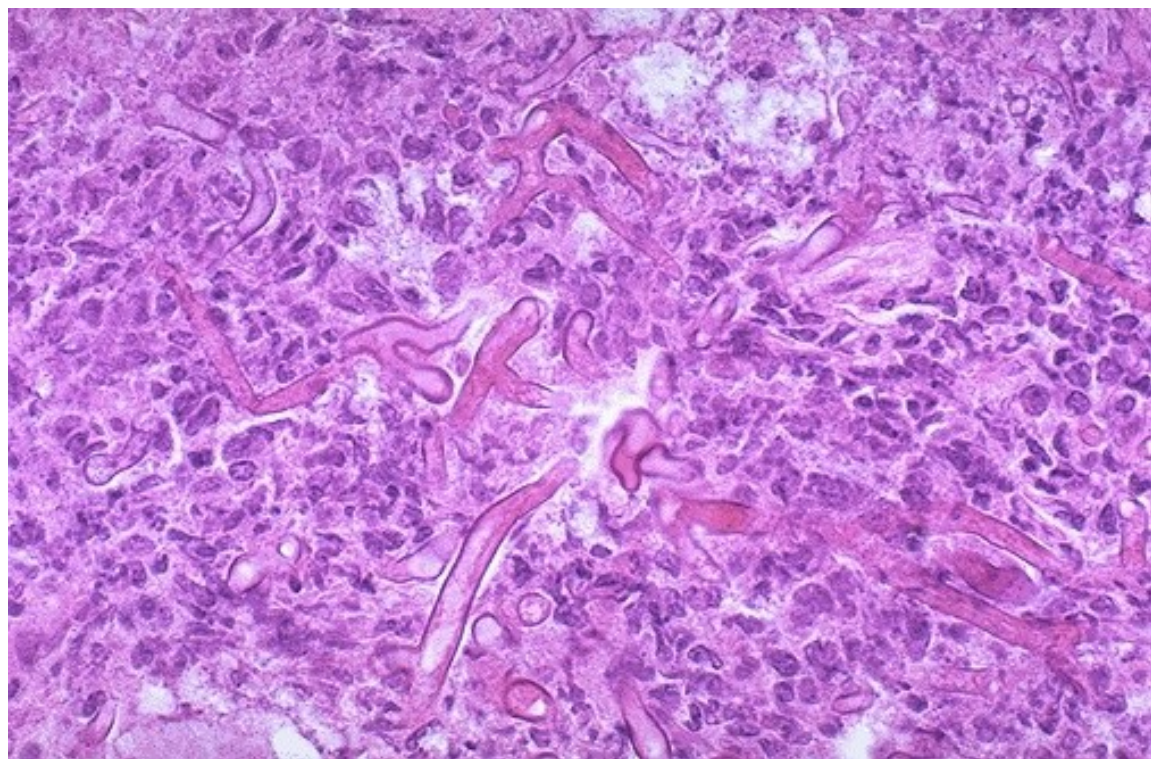
- The agents are most often inhabitants of the oral cavity
- Impairment of drainage of by edema of the mucosa may result in accumulation of pus - *empyema* of the sinus
- Accumulation of mucous secretions in the absence of bacterial invasion – *mucocele*

- Acute sinusitis may give rise to *chronic sinusitis*

- Uncommonly, sinusitis is a component of kartagener's syndrome which also includes bronchiectasis & situs invertus. These features occur secondary to defective ciliary action.

Necrotizing lesions of the nose & upper airways

- Mucormycosis - fungal infection particularly in diabetics
- Wegner granulomatosis



Nasopharynx

Pharyngitis & Tonsillitis

Causes

- ✓ concomitant with acute viral upper respiratory tract infections
- ✓ Bacteria can be superimposed on viral infections or bacteria can be the primary agent. The most common agents are β -hemolytic streptococci.

Morphologically, The inflammed nasopharyngeal mucosa may be reddened & in bacterial cases covered by exudative membrane & the nasopalatine & palatine tonsils may be enlarged & covered by exudates with reactive enlargement of the related lymphoid tissue.

- A typical appearance is of enlarged , reddened, tonsils dotted by pinpoints of exudates emanating from the tonsillar crypts – Follicular tonsillitis





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- The major importance of streptococcal pharyngitis is the late sequel – rheumatic fever & glomerulonephritis .
- Recurrent episodes of tonsillitis may lead to chronic tonsillitis.

- Recurrent attacks of tonsillitis may cause residual enlargement of the lymphoid tissue

Tumors of the nose, sinuses & nasopharynx

- Tumors are infrequent but include both mesenchymal & epithelial neoplasms
 - Nasopharyngeal angiofibroma
 - Sinonasal papilloma
 - Isolated plasmacytomas
 - Olfactory neuroblastoma
 - Nasopharyngeal carcinoma

Nasopharyngeal Carcinoma

- Three factors influence occurrence of this tumor

- Heredity

- Age

(the high frequency of this form of cancer in some parts of Africa in children & the Chinese adults which raises possibility of genetic susceptibility)

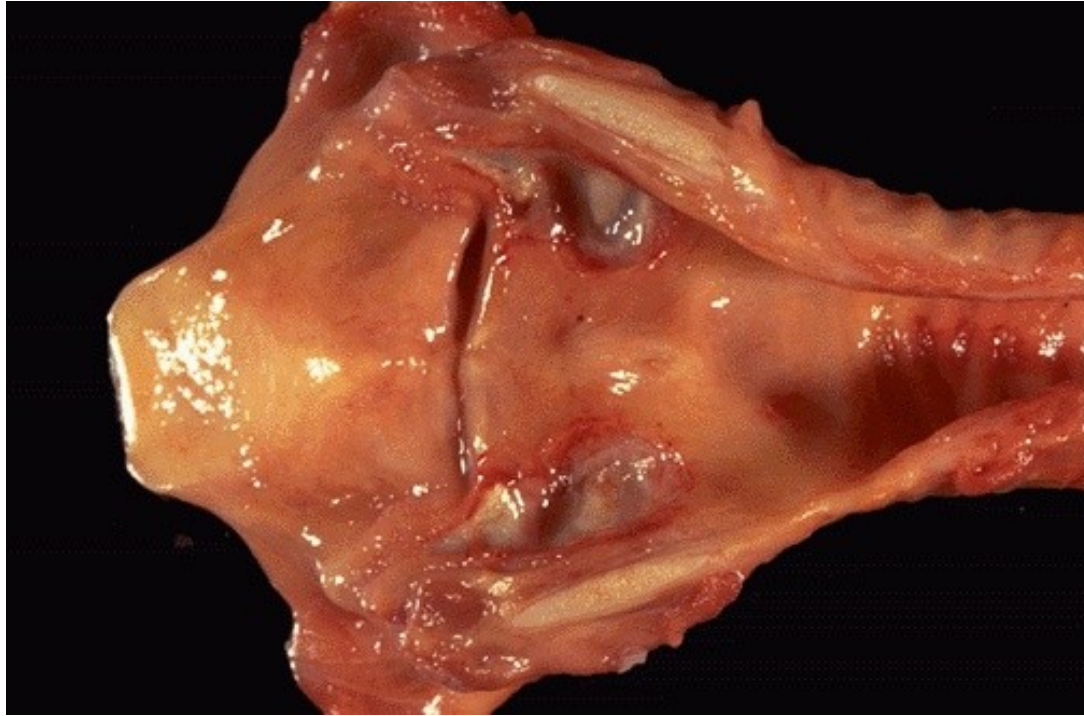
- **Infection with EBV** (EBV infects the host by first replicating in the nasopharyngeal epithelium and then infecting nearby tonsillar B lymphocytes. In some persons this leads to transformation of the epithelial cells.

Unlike the case with Burkitt lymphoma, another EBV-associated tumor, the EBV genome is found in virtually all nasopharyngeal carcinomas, including those that occur outside the endemic areas in Asia.

- It has three histologic patterns
 - ❖ Keratinizing squamous cell carcinoma
 - ❖ Non keratinizing squamous cell carcinoma
 - ❖ Undifferentiated carcinoma with abundant non epithelial lymphocytic infiltrate (lymphoepithelioma)

- Nasopharyngeal carcinoma tends to grow slowly & is often diagnosed after it has spread to regional cervical lymph nodes.
- Radiotherapy is the standard modality of treatment

Larynx

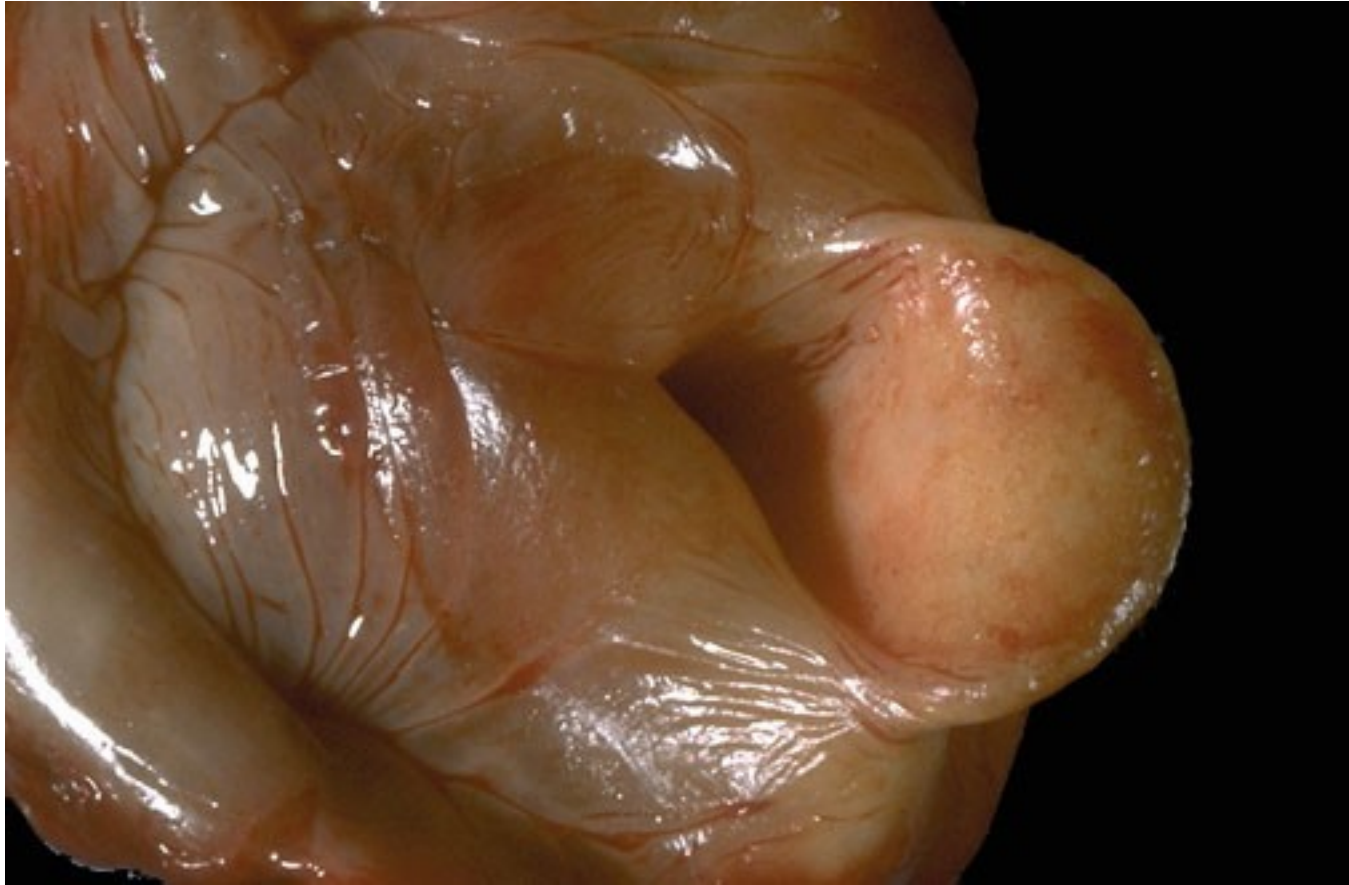


Inflammation

- Laryngitis may occur isolated due to viral, bacterial or chemical insult but more commonly it is part of upper respiratory tract infection or exposure to tobacco smoke.

- Most infections are self limited but it can be serious in infants & young children when mucosal congestion , exudation, or edema may cause laryngeal obstruction because of their small airways.

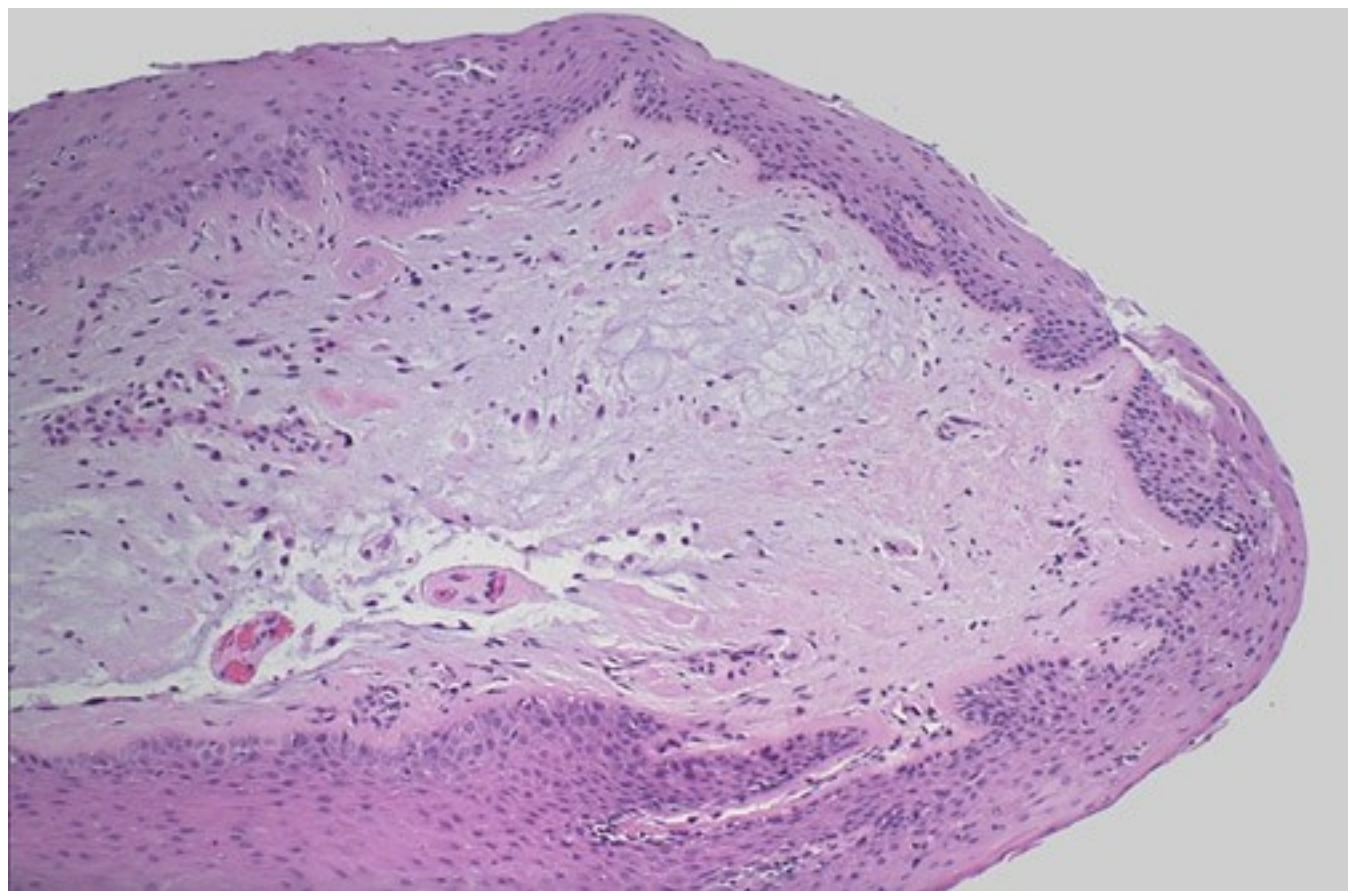
- Laryngioepiglottitis caused by *Haemophilus influenzae* or β - hemolytic streptococci in children may induce sudden swelling of epiglottis & vocal cords (Medical emergency).
- Croup is laryngotracheobronchitis in children in which inflammatory narrowing of the airways produces inspiratory stridor.

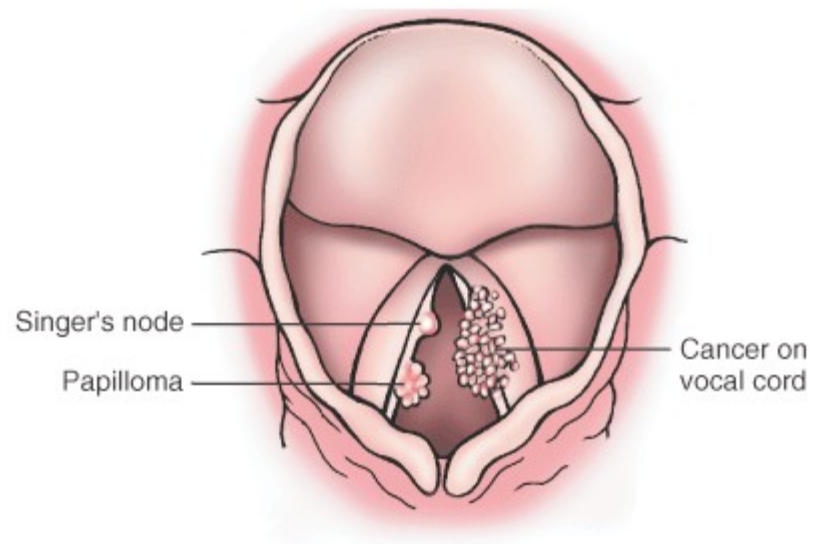


Reactive nodules (Vocal cord nodules ("polyps"))

- *Vocal cord nodules* ("polyps") are smooth, hemispherical protrusions (usually less than 0.5 cm in diameter) located, most often, on the true vocal cords.
- The nodules are composed of fibrous tissue and covered by stratified squamous mucosa that is usually intact but can be ulcerated by contact trauma with the other vocal cord.
- These lesions occur chiefly in heavy smokers or singers (singer's nodes), suggesting that they are the result of chronic irritation or abuse.

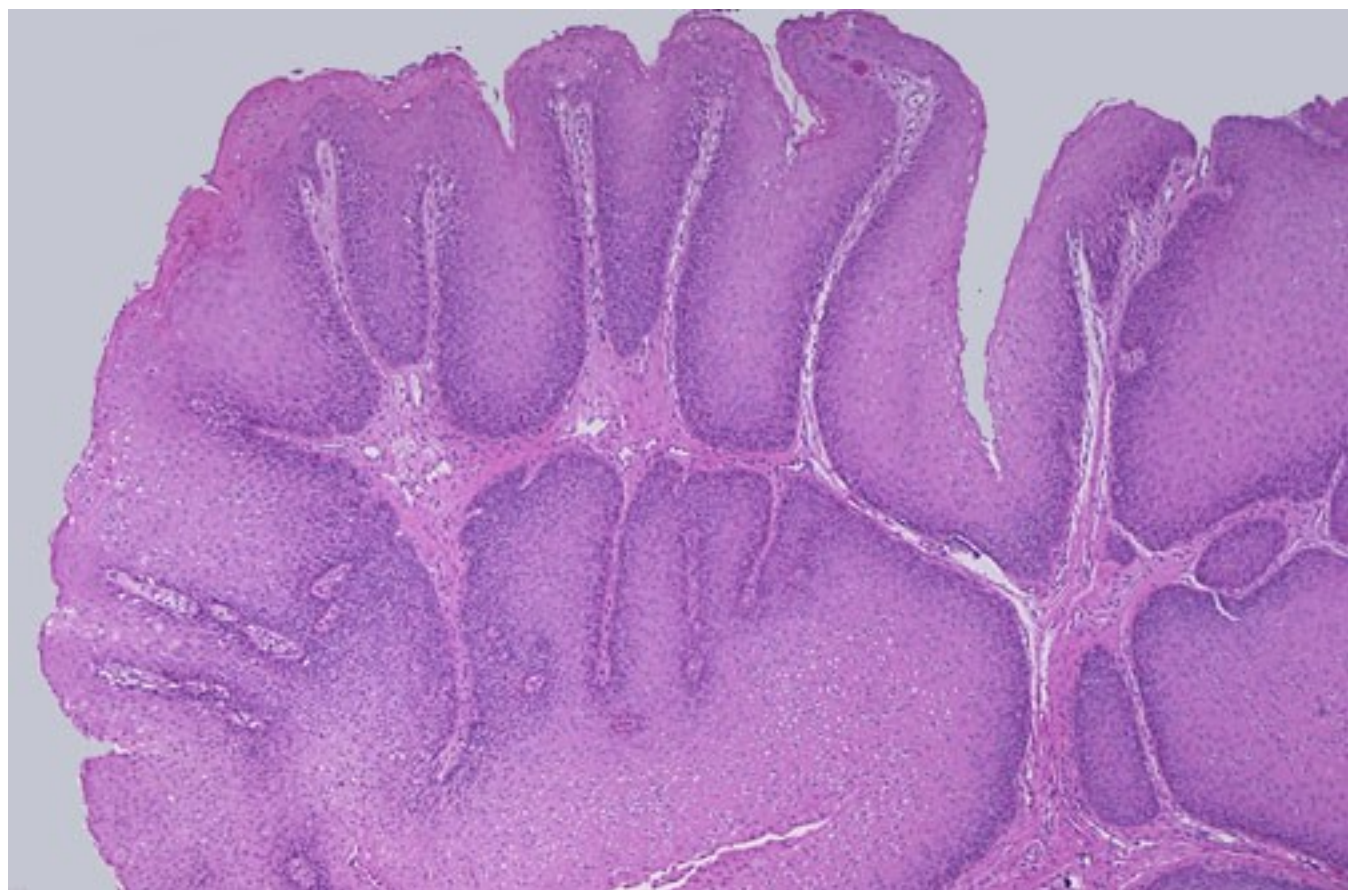






Laryngeal papilloma

- *Laryngeal papilloma* or *squamous papilloma* of the larynx is a benign neoplasm, usually on the true vocal cords, that forms a soft, raspberry-like excrescence rarely more than 1 cm in diameter.
- Histologically, it consists of multiple, slender, finger-like projections supported by central fibrovascular cores and covered by an orderly, typical, stratified squamous epithelium.
- These lesions are caused by human papillomavirus types 6 and 11, do not become malignant, but frequently recur.



CARCINOMA OF THE LARYNX

- Carcinoma of the larynx represents only 2% of all cancers.
- It most commonly occurs after age 40 years and is more common in men than in women.
- Nearly all cases occur in smokers, and alcohol and asbestos exposure may also play roles.

- Spectrum of epithelial alteration can be seen
larynx

Hyperplasia-atypical hyperplasia- dysplasia-
carcinoma in situ – Invasive carcinoma

- About 95% of laryngeal carcinomas are typical squamous cell tumors . Tumors more commonly arise on the vocal cords.
- Carcinoma of the larynx manifests itself clinically by persistent hoarseness



